TEXORA · COURSE NOTES · LECTURE SHEET

Multiple Integrals

Integration over areas and volumes — double and triple integrals, iterated evaluation, polar and other coordinate systems, and what they compute.

MATH-1101 · MATHEMATICS-I · TOPIC 5 · LEVEL 1 TERM 1

1. From lines to regions

An ordinary integral accumulates a function along a line segment. Real problems accumulate over areas and volumes instead: mass over a plate of varying density, volume under a curved surface, average temperature over a region. The **double integral** $\iint_R f(x, y)\, dA$ is defined exactly as before — divide the region R into tiny rectangles, multiply each area by the function's value there, add, and take the limit as the pieces shrink. When f = 1 it returns the area of R; when f is a density it returns total mass; when f is the height of a surface it returns the volume beneath it.

2. Iterated integrals: how double integrals are computed

The limit definition is never used directly. **Fubini's theorem** says the double integral over a rectangle equals an **iterated integral** — integrate with respect to one variable, then the other, in either order:

$$\iint_R f\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy.$$

WORKED EXAMPLE — A DOUBLE INTEGRAL OVER A RECTANGLE

Evaluate $\iint (x + y)\, dA$ over $R = [0, 1] \times [0, 2]$. Inner integral (over y): $\int_0^2 (x + y)\, dy = [xy + y^2/2]_0^2 = 2x + 2$. Outer integral: $\int_0^1 (2x + 2)\, dx = [x^2 + 2x]_0^1 = \mathbf{3}$.

Over a **general region**, the inner limits become functions: if R lies between $y = g_1(x)$ and $y = g_2(x)$ for $a \le x \le b$, the inner integral runs from $g_1(x)$ to $g_2(x)$ and the outer from a to b. Choosing the order of integration is a skill in itself — one order may give simple limits while the other forces the region to be split, and sometimes only one order can be evaluated in elementary functions at all.

WORKED EXAMPLE — VOLUME UNDER A PLANE

Volume under $z = 2 - x - y$ above the triangle $x \ge 0$, $y \ge 0$, $x + y \le 1$: $\int_0^1 \int_0^{1-x} (2 - x - y)\, dy\, dx$.

The inner integral gives $(1 - x)(3 - x)/2$; the outer gives $\tfrac{1}{2} \int_0^1 (3 - 4x + x^2)\, dx = \tfrac{1}{2}\,(3 - 2 + 1/3) =$

2/3.

3. Polar coordinates

For regions involving circles, the Cartesian grid fights the geometry; polar coordinates cooperate with it. With $x = r\cos\theta$, $y = r\sin\theta$, the area element becomes

$$dA = r\,dr\,d\theta$$

— the factor r (a Jacobian, met in Topic 4) accounts for the way polar rectangles widen as r grows. Forgetting this r is the single most common error in the topic.

> **WORKED EXAMPLE — AREA OF A CIRCLE, THE POLAR WAY**
>
> The disc of radius a is $0 \le r \le a$, $0 \le \theta \le 2\pi$. Area $= \int_0^{2\pi}\int_0^{a} r\,dr\,d\theta = \int_0^{2\pi} (a^2/2)\,d\theta = \mathbf{\pi a^2}$ — the familiar answer, obtained without a single square root.

4. Triple integrals

The **triple integral** $\iiint_V f(x, y, z)\,dV$ extends the idea to volumes: with $f = 1$ it gives the volume of V, and with f a density it gives mass. Evaluation is again by iteration — three nested ordinary integrals, with limits describing V slice by slice. Two coordinate systems often simplify the geometry: **cylindrical** coordinates (r, θ, z) with $dV = r\,dr\,d\theta\,dz$, for solids of revolution about an axis; and **spherical** coordinates (ρ, θ, φ) with $dV = \rho^2 \sin\varphi\,d\rho\,d\theta\,d\varphi$, for balls and spheres. Choosing coordinates that match the solid is the main strategic decision, exactly as with polar coordinates in the plane.

5. Applications at a glance

QUANTITY	INTEGRAL THAT GIVES IT
Area of region R	$\iint_R 1\,dA$
Volume under surface $z = f(x, y)$	$\iint_R f(x, y)\,dA$
Mass of a plate, density $\delta(x, y)$	$\iint_R \delta(x, y)\,dA$
Average value of f over R	$(1 / \text{area of R}) \cdot \iint_R f\,dA$
Volume / mass of a solid	$\iiint_V 1\,dV$ / $\iiint_V \delta\,dV$

Centre of mass, moments of inertia	Ratios of integrals of $x\cdot\delta$, $y\cdot\delta$, $(x^2 + y^2)\cdot\delta$ over the body

6. Changing the order of integration

A double integral stated in one order can often be re-stated in the other — the region is the same, only the slicing direction changes. Consider $\int_0^1 \int_y^1 f(x, y)\, dx\, dy$: the region is the triangle with vertices (0, 0), (1, 0), (1, 1), sliced horizontally. Sliced vertically instead, x runs from 0 to 1 and, for each x, y runs from 0 to x: the integral becomes $\int_0^1 \int_0^x f(x, y)\, dy\, dx$. The skill is entirely in the sketch: draw the boundary curves, mark which is the entrance and which the exit of a slice in each direction, and read the limits off the picture. When f contains a factor like e^{x^2} — integrable in only one order — this change is not a convenience but the only route to an answer.

7. Key points and summary

- Double and triple integrals are limits of sums over areas and volumes; Fubini’s theorem turns them into iterated ordinary integrals that can actually be computed.
- Over general regions the inner limits are functions; drawing the region and choosing the order of integration are the decisive steps.
- Changing coordinates costs a Jacobian factor: $dA = r\, dr\, d\theta$ in polar coordinates, $dV = r\, dr\, d\theta\, dz$ in cylindrical, $dV = \rho^2 \sin\varphi\, d\rho\, d\theta\, d\varphi$ in spherical. Match the coordinates to the geometry.
- With f = 1 these integrals return areas and volumes; with f a density they return mass, and their ratios give averages, centres of mass and moments of inertia.

8. Practice

Work each problem fully; final answers are given at the end.

1. Evaluate $\iint (x - y)\, dA$ over the unit square $[0, 1] \times [0, 1]$.
2. Evaluate $\iint xy\, dA$ over $[0, 2] \times [0, 1]$ (try separating the integrals).
3. Find the area enclosed between $y = x$ and $y = x^2$.
4. Using polar coordinates, find the area of a semicircle of radius 2.
5. Evaluate $\iiint xyz\, dV$ over the unit cube $[0, 1] \times [0, 1] \times [0, 1]$.

Answers

1. 0 — the positive and negative halves cancel by symmetry.
2. $(\int_0^2 x\, dx)(\int_0^1 y\, dy) = (2)(1/2) = 1$.

3. $\int_0^1 (x - x^2)\,dx = 1/2 - 1/3 = 1/6$.

4. $\int_0^{\pi} \int_0^2 r\,dr\,d\theta = \pi \cdot 2 = 2\pi$ (half of $\pi \cdot 2^2$, as expected).

5. $(\int_0^1 x\,dx)^3 = (1/2)^3 = 1/8$.
